

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2023

Bachelor in Computer Applications

Course Title: **Cloud Computing**

Code No: **CACS402** Semester: **VII**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. Compare and contrast the cloud service model.
3. How does SaaS differ from other types of software delivery models?
4. What are the main benefits of using a Community Cloud, and what types of organizations or industries are best suited for this type of deployment model?
5. Give a brief description of the organization's cloud adaptation roadmap.
6. What are the main roles within a Service-oriented Architecture (SOA) and how do they interact and collaborate to deliver value to the business?
7. How is the Aneka cloud platform designed to manage distributed computing, and what is its core architecture?
8. Write short notes on (Any Two): a) Public and Private Cloud b) IAAS c) Data Security

Group C

Attempt any TWO questions [2x10=20]

9. Explain the concept of cloud security and the measures that can be taken to secure cloud-based applications and services. Additionally, identify the key threats to cloud security and explain how they can be mitigated.
10. Describe the concept behind MapReduce, its function in distributed computing, the procedures required in a MapReduce operation, and give a practical example.
11. Discuss the major elements of a virtualization techniques, such as hypervisors and virtual machines, and compare and contrast them. You should also discuss the benefits and drawbacks of each technology.